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Full Extracting Kinematic and Dynamic Equations of X/Y Pedestal with Velocity Analysis

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***Abstract*—In order to track the satellite, it is necessary to extract the pedestal kinematic equations by assuming to have the azimuth/elevation angles of satellite path. This paper deals with extracting of forward and inverse kinematic equations of the x/y pedestal with analysis of speed and extraction of singular points. Then equations are varied by the real Azimuth/Elevation of a LEO satellite. Finally the velocity and input torque of each link are examined during the period of work.**

Keywords-Pedestal; Azimuth/elevation Angles; Forward and Inverse Kinematic;dynamic,Singular points; Velocity

I. INTRODUCTION

Nowadays different kinds of pedestal are made and designed to tracking the satellite in different frequency bands that the most noticeable of them is 2 axes azimuth/elevation pedestal in recent decades. 2-DOF pedestals are the best because they have less movable pieces and less problems and accessories, in addition to previous advantages its servo and control system are much simple. In the late of fifties and early sixties, the first smart LEO satellite was launched it showed a real main limitation of azimuth/elevation pedestal. This limitation occurs while the satellite passes exactly above the pedestal because in this situation pedestal must reach to infinity velocity in elevation axes to tracking satellite continuously. To solve this problem x/y pedestal was used. In the late 1990s x/y pedestal was designed instead of previous ones. The new one has these following advantages [1]:

a. Ability to track all targets with azimuth angle above 5 degree, Zenith area is exactly above the pedestal and can moved from 90 degrees zenith to two locations on the horizon. Since satellites are tracked with an azimuth angle above 10 degree, this pedestal can tracks all LEO satellites.
b. Reduction of requirement torque in comparison to previous plants.
c. Less Speed and acceleration.
d. Mechanical stability of motors and tracking accuracy increase due to the low speed of motors.

For tracking the satellite, extracting of pedestal kinematic equations by assumption of having azimuth/elevation angles of satellite path is necessary, finally, by solving the inverse kinematic the pedestal joint angles achieve as input to the controller unit. The concept of forward kinematic is widely used specially in robot manipulators [2, 3, 4]. In [5] the kinematic of ship mounted satellite tracking antenna is extracted.

Inverse kinematics problem is one of the fundamental problems. This problem consists in defining a motion of pedestal links, able to produce a desirable motion. Practical inverse kinematics rely on the manipulator Jacobian maintains its full rank. It is a much more difficult problem than forward kinematics. It has a wide range of application as [6, 7]. The different techniques used for solving inverse kinematics can be classified as algebraic [8-12], geometric [9, 13, 14], and iterative [10]. In this paper we will restrict our attention to algebraic. There are some values where the Jacobian becomes singular. Such locations are called singularities. In singular configuration the rigid body lost one or more degrees of freedom. These are very important for design, trajectory planning, and control. Inverse kinematic is based on the joint velocities which are linearly related to the desired end-effector velocity by the time-varying Jacobian matrix. These concepts to analyze the motion are important.

After exact and scrutinized studies in the field of pedestals, the authors assume that complete extracting of forward/inverse kinematic equation with velocity analysis are presented in the first place.

This paper is organized as follows, in section II x/y pedestal forward/inverse kinematic is extracted with velocity analysis and singular points in section III. In section IV accuracy of obtained kinematic equation is studied and velocity analysis of each link is done by inverse dynamic

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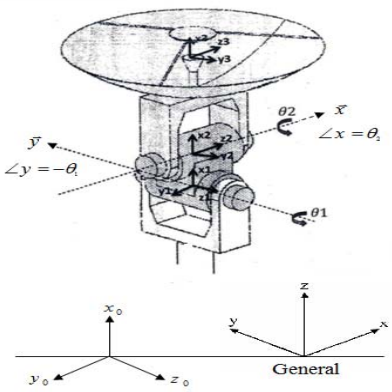

Fig. 1. x/y pedestal with coordinate axis to calculate of kinematic equation

controller over a real path of satellite. Finally in section V conclusions are presented.

II. KINEMATIC EQUATIONS OF X/Y PEDESTAL

A. Forward Kinematic

X/Y pedestal uniquely suited to LEO satellite is shown in Fig.1. It is designed such that the two driven axes intersect at a common point behind the reflector. The X axis is oriented north-south and is parallel to the surface of earth, the X angle is taken to be zero when the east component of range is zero, limits of +/- 90 degrees are set on the value of this angle, and the sign of angle is taken to be the same as the sign of the east component. The Y angle is taken to be zero when the north component of range is zero, limits of +/- 90 degrees are set on the value of the angle, and its sign is taken to be the same as sign of the north component [1].

Kinematic is the science of motion which treats motion without regard to the forces that cause it. To analysis of kinematic, we set up a system of coordinate frames. This system allows one to relate the coordinates of any frame relative to any other frame by means of coordinate transformation. We attach a frame to the base of the mechanism, called frame {0}. Frame {1} is coincident with link 1, Frame {2} is coincident with link 2 and Frame {3} is coincident with antenna. Frame {G} is reference frame. Note that all joints are rotational. We determine the transform which defines frame {i} relative to the frame {i-1}. In general, this transformation will be a function of the four link parameters [9].

$$ {}^{i-1}_{i}T = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ s\theta_i c\alpha_{i-1} & c\theta_i c\alpha_{i-1} & -s\alpha_{i-1} & -s\alpha_{i-1}.d_i \\ s\theta_i s\alpha_{i-1} & c\theta_i s\alpha_{i-1} & c\alpha_{i-1} & -c\alpha_{i-1}.d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (1) $$

In the above equation $c = \cos$, $s = \sin$, and the following definitions of the link parameters are valid:

a_i = the distance from z_i to z_{i+1} measured along x_i.

α_i =the angle between z_i to z_{i+1} measured along x_i.

d_i =the distance from x_{i-1} to x_i measured along z_i.

θ_i = the angle between x_{i-1} to x_i measured along z_i.

Transformation matrix that relates frame {1} to frame {0} is given by:

$$ {}^{0}_{1}T = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & h_0 \\ s\theta_1 & c\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (2) $$

Rotation matrix that relates frame {1} to frame {0} is given by:

$$ {}^{0}_{1}R = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 \\ s\theta_1 & c\theta_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (3) $$

Transformation and rotation matrix that relates frame {2} to frame{1} and Transformation and rotation matrix that relates frame {3} to frame{2} are given by:

$$ {}^{1}_{2}T = \begin{bmatrix} c\theta_2 & -s\theta_2 & 0 & h_1 \\ 0 & 0 & -1 & 0 \\ s\theta_2 & c\theta_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (4) $$

$$ {}^{2}_{1}R = {}^{1}_{2}R^T = \begin{bmatrix} c\theta_2 & 0 & s\theta_2 \\ -s\theta_2 & 0 & c\theta_2 \\ 0 & -1 & 0 \end{bmatrix} $$

$$ {}^{2}_{3}T = \begin{bmatrix} 1 & 0 & 0 & h_2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \Rightarrow {}^{3}_{2}R = {}^{2}_{3}R^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} $$

Rotation and transformation matrix that relates frame {0} to frame {G} are given by:

$$ {}^{G}_{0}T = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \; {}^{G}_{0}R = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \quad (5) $$

Then, the link transformation can be multiplied together to find the single transformation that relate frame {3} to frame {0}:

$$ {}^{0}_{3}T = {}^{0}_{1}T \times {}^{1}_{2}T \times {}^{2}_{3}T $$

$$ {}^{0}_{3}T = \begin{bmatrix} c\theta_1 c\theta_2 & -c\theta_1 s\theta_2 & s\theta_1 & h_2 c\theta_1 c\theta_2 + h_1 c\theta_1 + h_0 \\ s\theta_1 c\theta_2 & -s\theta_1 s\theta_2 & -c\theta_1 & h_1 s\theta_1 + h_2 s\theta_1 c\theta_2 \\ s\theta_2 & c\theta_2 & 0 & h_2 s\theta_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{6} $$

And the rotation matrix:

$$ {}^{0}_{3}R = \begin{bmatrix} c\theta_1 c\theta_2 & -c\theta_1 s\theta_2 & s\theta_1 \\ s\theta_1 c\theta_2 & -s\theta_1 s\theta_2 & -c\theta_1 \\ s\theta_2 & c\theta_2 & 0 \end{bmatrix} \tag{7} $$

Also,

$$ {}^{G}_{3}R = {}^{G}_{0}R \; {}^{0}_{3}R = \begin{bmatrix} s\theta_1 c\theta_2 & -s\theta_1 s\theta_2 & -c\theta_1 \\ s\theta_2 & c\theta_2 & 0 \\ c\theta_1 c\theta_2 & -c\theta_1 s\theta_2 & s\theta_1 \end{bmatrix} \tag{8} $$

Displacement vector will be equal to:

$$ p = \begin{bmatrix} s\theta_1 . c\theta_2 . h_2 + s\theta_1 . h_1 \\ s\theta_2 . h_2 \\ c\theta_1 . c\theta_2 . h_2 + c\theta_1 . h_1 + h_0 \end{bmatrix} \tag{9} $$

B. Inverse Kinematic

In this part, we compute the set of joint angles which allow antenna to operate in horizontal (Azimuth) and vertical plane (Elevation). Based on Fig.2, unit vector $\vec{P}$:

$$ \vec{P} = \begin{bmatrix} -\cos(Az).\cos(Elv) \\ \sin(Az).\cos(Elv) \\ \sin(Elv) \end{bmatrix} \tag{10} $$

To tracking satellite accurately, axis which is perpendicular to the plate of the antenna (antenna line of sight) must be put in the same direction with unit vector $\vec{P}$ so we have:

$$ {}^{G}_{3}\vec{P} = {}^{G}_{3}R.\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad , \quad {}^{G}_{3}R.\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \sin\theta_1 \cos\theta_2 \\ \sin\theta_2 \\ \cos\theta_1 \cos\theta_2 \end{bmatrix} \tag{11} $$

$$ \Rightarrow \sin(Az).\cos(Elv) = \sin\theta_2 $$

$$ \Rightarrow \theta_2 = \sin^{-1}[\sin(Az).\cos(Elv)] $$

$$ \left.\begin{aligned} -\cos(Az).\cos(Elv) &= \sin\theta_1 \cos\theta_2 \\ \sin(Elv) &= \cos\theta_1 \cos\theta_2 \end{aligned}\right\} $$

$$ \Rightarrow \theta_1 = -A\tan\left[\frac{\cos(Az).\cos(Elv)}{\sin(Elv)}\right] $$

$$ -\frac{\pi}{2} < \theta_1 \;,\; \theta_2 < +\frac{\pi}{2} $$

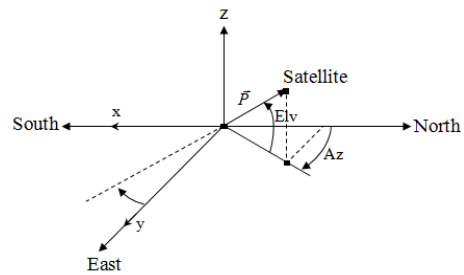

Fig. 2. Satellite Azimuth/Elevation angles in ground station

To have angles in trigonometric we suppose using of Atan2 instead of Atan:

$$ \mathrm{Atan2}(y,x) = \begin{cases} \mathrm{Atan}(y/x) & \text{if } x > 0 \\ \mathrm{sign}(y)(\pi - \mathrm{Atan}(|y/x|)) & \text{if } x < 0 \\ 0 & \text{if } x = y = 0 \\ \mathrm{sign}(y)\pi/2 & \text{if } x = 0, y \neq 0 \end{cases} \tag{12} $$

B .Dynamic

We need the pedestal dynamic model for system control and determination of some pedestal practical characteristics as required input torques.

Dynamic equations are calculated as the differential equations in a joint space. The Euler-Lagrange method is used to extract dynamic equations of the pedestal. In this method, the kinetic and potential energy of the system are calculated.

The total kinetic and potential energy of the pedestal mechanism are equal to [14, 9]:

$$ \begin{cases} k_i = \frac{1}{2} m_i \, v_{c_i}^T . v_{c_i} + \frac{1}{2} \, {}^{i}w_i^T . {}^{c_i}I_i . {}^{i}w_i \\ \Rightarrow K = \sum k_i \\ u_i = -m_i . {}^{g}g^T . {}^{g}p_{c_i} + u_{ref_i} \\ \Rightarrow U = \sum u_i \end{cases} \tag{13} $$

We know Lagrangian function is as follow:

$$ L(\theta, \dot{\theta}) = K(\underline{\theta}, \underline{\dot{\theta}}) - U(\underline{\theta}) \tag{14} $$

According to The Lagrange equation (Eq.14), motion equations are obtained as follows:

$$ \frac{d}{dt}\left(\frac{\partial L}{\partial \underline{\dot{\theta}}}\right) - \frac{\partial L}{\partial \underline{\theta}} = \tau \tag{15} $$

$$ \frac{d}{dt} . \frac{\partial K(\underline{\theta}, \underline{\dot{\theta}})}{\partial \underline{\dot{\theta}}} - \frac{\partial(\underline{\theta}, \underline{\dot{\theta}})}{\partial \theta} + \frac{\partial U(\underline{\theta})}{\partial \theta} = \underline{\tau} $$

Where, $\underline{\tau} = \begin{pmatrix} \tau_1 \\ \tau_2 \end{pmatrix}$, $\underline{\theta} = \begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix}, \underline{\dot{\theta}} = \begin{pmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{pmatrix}$, Kinetic energy of the base link is $k_0 = 0$ and kinetic energy of link1and 2 are

$$k_1 = \tfrac{1}{2} m_1 . v_{c_1}^T . v_{c_1} + \tfrac{1}{2}\, {}^1w_1^T . I_1 . {}^1w_1 \quad ;$$

$${}^1w_1 = \begin{bmatrix} 0 & 0 & \dot{\theta}_1 \end{bmatrix}^T ; {}^1v_{c_1} = {}^1v_1 + {}^1w_1 \times {}^1p_{c_1} \; ; \; {}^1p_{c_1} = \begin{bmatrix} x_{c_1} & 0 & 0 \end{bmatrix}^T$$

$$= \begin{bmatrix} 0 & x_{c_1} . \dot{\theta}_1 & 0 \end{bmatrix}^T ; {}^1v_1 = [0 \quad 0 \quad 0]$$

$$\Rightarrow k_1 = \tfrac{1}{2} m_1 x_{c_1}^2 \dot{\theta}_1^2 + \tfrac{1}{2} . I_{zz_1} . \dot{\theta}_1^2 \; , \; I_1 = \begin{bmatrix} I_{xx_1} & 0 & 0 \\ 0 & I_{yy_1} & 0 \\ 0 & 0 & I_{zz_1} \end{bmatrix} \tag{16}$$

$$k_2 = \tfrac{1}{2} m_2 . v_{c_2}^T . v_{c_2} + \tfrac{1}{2}\, {}^2w_2^T . I_2 . {}^2w_2 \tag{17}$$

$${}^2w_2 = \begin{bmatrix} \sin\theta_2 . \dot{\theta}_1 & \cos\theta_2 . \dot{\theta}_1 & \dot{\theta}_2 \end{bmatrix}$$

$${}^2v_{c_2} = {}^2v_2 + {}^2w_2 \times {}^2p_{c_2} \; ; \; {}^2p_{c_2} = \begin{bmatrix} x_{c_2} & 0 & 0 \end{bmatrix}^T$$

$$= \begin{bmatrix} 0 \\ x_{c_2} . \dot{\theta}_2 \\ x_{c_2} . \cos\theta_2 . \dot{\theta}_1 - h_1 . \dot{\theta}_1 \end{bmatrix} ; {}^2v_2 = \begin{bmatrix} 0 \\ 0 \\ h_1 . \dot{\theta}_1 \end{bmatrix}$$

$$\begin{aligned} k_2 &= \tfrac{1}{2} m_2 \times \\ &\left[x_{c_2} . \dot{\theta}_2^2 + x_{c_2} \cos^2\theta_2 . \dot{\theta}_1^2 + h_1^2 . \dot{\theta}_1^2 + 2 h_1 x_{c_2} \cos\theta_2 . \dot{\theta}_1^2 \right] \\ &+ \tfrac{1}{2} \left[I_{xx2} \sin^2\theta_2 . \dot{\theta}_1^2 + I_{yy2} \cos^2\theta_2 . \dot{\theta}_1^2 + I_{zz2} \dot{\theta}_2^2 \right] \\ &= \dot{\theta}_1^2 \Big(\tfrac{1}{2} m_2 x_{c_2}^2 \cos^2\theta_2 + \tfrac{1}{2} m_2 h_1^2 + \\ &\quad \tfrac{1}{2} I_{xx2} \sin^2\theta_2 + \tfrac{1}{2} I_{yy2} \cos^2\theta_2 + \\ &\quad m_2 h_2 x_{c_2} \cos\theta_2 \Big) + \\ &\quad \left(\tfrac{1}{2} m_2 x_{c_2}^2 + \tfrac{1}{2} I_{zz2} \right) \dot{\theta}_2^2 \end{aligned} \tag{18}$$

Total kinetic energy of mechanisms:

$$\Rightarrow k = k_0 + k_1 + k_2 \quad \Rightarrow k = 0 + k_1 + k_2 = k_1 + k_2 \tag{19}$$

By substituting above equation in Eq.15 we have:

$$\frac{d}{dt} . \frac{\partial K}{\partial \dot{\theta}} = \frac{d}{dt} . \begin{bmatrix} 2\Big(\tfrac{1}{2} m_2 x_{c_2}^2 \cos^2\theta_2 + \tfrac{1}{2} m_2 h_1^2 + \\ \tfrac{1}{2} I_{xx2} \sin^2\theta_2 + \tfrac{1}{2} I_{yy2} \cos^2\theta_2 + \\ m_2 h_1 x_{c_2} \cos\theta_2 + \tfrac{1}{2} m_1 x_{c_1}^2 + \tfrac{1}{2} I_{zz1} \Big) \dot{\theta}_1 \\ 2\left(\tfrac{1}{2} m_2 x_{c_2}^2 + \tfrac{1}{2} I_{zz2} \right) \dot{\theta}_2 \end{bmatrix} \tag{20}$$

$$= \begin{bmatrix} \Big(m_2 x_{c_2}^2 \cos^2\theta_2 + m_2 h_1^2 + I_{xx2} \sin^2\theta_2 + \\ I_{yy2} \cos^2\theta_2 + 2 . m_2 h_1 x_{c_2} \cos\theta_2 + m_1 x_{c_1}^2 + I_{zz1} \Big) \ddot{\theta}_1 + \\ \Big(-2 m_2 x_{c_2}^2 \sin\theta_2 \cos\theta_2 + 2 I_{xx2} \cos\theta_2 \sin\theta_2 - \\ 2 I_{yy2} \sin\theta_2 \cos\theta_2 - m_2 h_1 x_{c_2} \sin\theta_2 \Big) \dot{\theta}_1 \dot{\theta}_2 \\ (m_2 x_{c_2}^2 + I_{zz2}) \ddot{\theta}_2 \end{bmatrix}$$

$$\frac{\partial K}{\partial \theta} = \begin{bmatrix} 0 \\ \dot{\theta}_1^2 \Big(-m_2 x_{c_2}^2 \sin\theta_2 \cos\theta_2 + I_{xx2} \cos\theta_2 \sin\theta_2 - \\ I_{yy2} \sin\theta_2 \cos\theta_2 - m_2 h_1 x_{c_2} \sin\theta_2 \Big) \end{bmatrix} \tag{21}$$

$$\frac{\partial U}{\partial \theta} = \begin{bmatrix} -g m_1 x_{c_1} \sin\theta_1 - g m_2 h_1 \sin\theta_1 \\ -g m_2 x_{c_2} \sin\theta_1 \cos\theta_2 \\ -g m_2 x_{c_2} \cos\theta_1 \sin\theta_2 \end{bmatrix}$$

Finally, the system dynamic equation is obtained as:

$$\left[M(\underline{\theta}) \right] \begin{bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{bmatrix} + \left[C(\underline{\theta}, \underline{\dot{\theta}}) \right] + \left[G(\underline{\theta}) \right] = \begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix} \tag{22}$$

$$M(\underline{\theta}) = \begin{bmatrix} m_2 x_{c_2}^2 \cos^2\theta_2 + m_2 h_1^2 + I_{xx2}\sin^2\theta_2 + I_{yy2}\cos^2\theta_2 + 2.m_2 h_1 x_{c_2}\cos\theta_2 + m_1 x_{c_2}^2 + I_{zz1} & 0 \\ 0 & m_2 x_{c_2}^2 + I_{zz2} \end{bmatrix}$$

$$C(\underline{\theta},\underline{\dot{\theta}}) = \begin{bmatrix} \left(-2m_2 x_{c_2}^2 \sin\theta_2\cos\theta_2 + 2 I_{xx2}\cos\theta_2\sin\theta_2 - 2 I_{yy2}\sin\theta_2\cos\theta_2 - m_2 h_1 x_{c_2}\sin\theta_2\right)\dot{\theta}_1\dot{\theta}_2 \\ \left(-m_2 x_{c_2}^2 \sin\theta_2\cos\theta_2 + I_{xx2}\cos\theta_2\sin\theta_2 - I_{yy2}\sin\theta_2\cos\theta_2 - m_2 h_1 x_{c_2}\sin\theta_2\right)\dot{\theta}_1^2 \end{bmatrix}$$

$$G(\underline{\theta}) = \begin{bmatrix} -m_1 g x_{c_1}\sin\theta_1 - m_2 g\left(h_1\sin\theta_1 + x_{c_2}\sin\theta_1\cos\theta_2\right) \\ -m_2 g x_{c_2}\cos\theta_1\sin\theta_2 \end{bmatrix}$$

τ_1 is input torque for axis y and link1 , τ_2 is input torque for axis x and link2.

III. VELOCITY ANALYSIS AND SINGULARITIETS

A. *Velocity Analysis*

Study of both velocities and forces lead to a matrix entity called Jacobian [9].

In considering the motion of links we will always use link frame {0} as our reference frame. Hence, v_i is the linear velocity of the origin of link frame {i}, and ω_i is the angular velocity of link frame {i}. At any time, each link in motion has some linear and angular velocity. We compute the velocities of each link in order starting from the base. So in frame {i}:

$$^{i+1}\omega_{i+1} = {}^{i+1}_{i}R\,{}^{i}\omega_i + \dot{\theta}_{i+1}\,{}^{i+1}\hat{Z}_{i+1} \quad ,i = 0,1,2 \tag{23}$$

$$^{i+1}v_{i+1} = {}^{i+1}_{i}R\,({}^{i}v_i + {}^{i}\omega_i \times {}^{i}P_{i+1}) \quad ,i = 0,1,2 \tag{24}$$

Because base is constant therefore:

$$^{0}\omega_0 = 0 \quad , {}^{0}v_0 = 0 \tag{25}$$

By using:

$$^{1}p_2 = \begin{bmatrix} h_1 \\ 0 \\ 0 \end{bmatrix} \quad , {}^{2}p_3 = \begin{bmatrix} h_2 \\ 0 \\ 0 \end{bmatrix} \tag{26}$$

We obtain linear and angular velocity for x/y pedestal:

$$^{1}\omega_1 = \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix} \quad , {}^{1}v_1 = 0 \tag{27}$$

$$^{2}\omega_2 = \begin{bmatrix} \dot{\theta}_1 S\theta_2 \\ \dot{\theta}_1 C\theta_2 \\ \dot{\theta}_2 \end{bmatrix} \quad , {}^{2}v_2 = \begin{bmatrix} 0 \\ 0 \\ -\dot{\theta}_1 h_1 \end{bmatrix}$$

$$^{3}\omega_3 = \begin{bmatrix} \dot{\theta}_1 S\theta_2 \\ \dot{\theta}_1 C\theta_2 \\ \dot{\theta}_2 \end{bmatrix} \quad , {}^{3}v_3 = \begin{bmatrix} 0 \\ \dot{\theta}_2 h_2 \\ -\dot{\theta}_1 h_1 - h_2\dot{\theta}_1 C\theta_2 \end{bmatrix}$$

We generally use Jacobians which relate joint velocities to Cartesian velocities of x/y pedestal.

$$V = J(\theta)\,\dot{\theta} \tag{28}$$

Where, θ is the vector of joint angles and V is a vector of Cartesian velocities.
Cartesian velocity vector is the linear velocity vector and rotational velocity vector stacked together:

$$V = \begin{bmatrix} v \\ \omega \end{bmatrix} \tag{29}$$

Cartesian velocity vector in frame {3} is given by:

$$^{3}V = \begin{bmatrix} v \\ \omega \end{bmatrix} = \begin{bmatrix} 0 \\ \dot{\theta}_2 h_2 \\ -\dot{\theta}_1 h_1 - h_2\dot{\theta}_1 C\theta_2 \\ \dot{\theta}_1 S\theta_2 \\ \dot{\theta}_1 C\theta_2 \\ \dot{\theta}_2 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 0 \\ 0 & h_2 \\ -h_1 - h_2 C\theta_2 & 0 \\ S\theta_2 & 0 \\ C\theta_2 & 0 \\ 0 & 1 \end{bmatrix}}_{J(\theta)} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} \tag{30}$$

Then, we obtain the Jacobain matrix which relates joint angles to AZ/EL angles. Based on Fig.2:

$$^{G}_{3}R_{y,z}(-El, -\pi - Az) = R_z(-\pi - Az)R_y(-El) \tag{31}$$

$$= \begin{bmatrix} -\cos Az & -\sin Az & 0 \\ \sin Az & -\cos Az & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos El & 0 & -\sin El \\ 0 & 1 & 0 \\ \sin El & 0 & \cos El \end{bmatrix}$$

$$= \begin{bmatrix} -\cos Az\cos El & -\sin Az & \cos Az\sin El \\ \sin Az\cos El & -\cos Az & -\sin Az\sin El \\ \sin El & 0 & \cos El \end{bmatrix}$$

This rotation matrix is a proper orthogonal. There is a skew-symmetric matrix(S matrix) for each proper orthogonal as:

$$S = \begin{bmatrix} 0 & -\Omega_z & \Omega_y \\ \Omega_z & 0 & -\Omega_x \\ -\Omega_y & \Omega_x & 0 \end{bmatrix} = \dot{R}R^{-1} \tag{32}$$

$$\Omega = \begin{bmatrix} \Omega_x \\ \Omega_y \\ \Omega_z \end{bmatrix}, R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

$$\Omega_x = \dot{r}_{31}r_{21} + \dot{r}_{32}r_{22} + \dot{r}_{33}r_{23}$$
$$\Omega_y = \dot{r}_{11}r_{31} + \dot{r}_{12}r_{32} + \dot{r}_{13}r_{33}$$
$$\Omega_z = \dot{r}_{21}r_{11} + \dot{r}_{22}r_{12} + \dot{r}_{23}r_{13}$$

Where, vector Ω is angular velocity vector.
Consider a fixed vector P with in frame {B}, which is rotating, it is description in another frame {A} is given as:

$$^{A}V_P = {}^{A}\Omega_B \times {}^{A}P \tag{33}$$

Where, $^{A}\Omega_B$ is angular velocity vector which specifies the motion of frame {B} with respect to frame {A}. Making use of Eq.31, we have:

$$\left.\begin{aligned} \Omega_x &= \sin Az\, \dot{El} \\ \Omega_y &= \cos Az\, \dot{El} \\ \Omega_z &= \dot{Az} \end{aligned}\right\} \Rightarrow \Omega = \begin{bmatrix} \sin Az & 0 \\ \cos Az & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{El} \\ \dot{Az} \end{bmatrix} \tag{34}$$

Based on Eq.8

$$\left.\begin{aligned} \Omega_x &= \dot{\theta}_1 \\ \Omega_y &= \cos\theta_1\, \dot{\theta}_2 \\ \Omega_z &= \sin\theta_1\, \dot{\theta}_2 \end{aligned}\right\} \Rightarrow \Omega = \begin{bmatrix} 1 & 0 \\ 0 & \cos\theta_1 \\ 0 & \sin\theta_1 \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} \tag{35}$$

It is easily verified that:

$$\begin{bmatrix} \dot{El} \\ \dot{Az} \end{bmatrix} = \begin{bmatrix} \sin Az & \cos Az \cos\theta_1 \\ 0 & \sin\theta_1 \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} \tag{36}$$

And by replacing Eq.11 to Eq.36

$$\begin{bmatrix} \dot{El} \\ \dot{Az} \end{bmatrix} = \underbrace{\begin{bmatrix} \dfrac{-\sin\theta_1 \cos\theta_2}{\sqrt{1-\cos^2\theta_2\cos^2\theta_1}} & \dfrac{-\sin\theta_2}{\sqrt{1-\cos^2\theta_2\cos^2\theta_1}} \cos\theta_1 \\ 0 & \sin\theta_1 \end{bmatrix}}_{J(\theta)} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} \tag{37}$$

B. *Singularities*

There are values, where the Jacobian becomes singular. Such locations are called singularities. All mechanisms have singularities at the boundary of their workspace, and most have loci of singularities inside their workspace.

When a system is in a singular configuration, it has lost one or more degree of freedom. This means that there is some direction in Cartesian space which is impossible to move. To find the singular point we must equal the determinant of Jacobian to zero. So singular points of x/y pedestal are:

$$|J(\theta)| = 0 \Rightarrow \frac{-\sin^2\theta_1 \cos\theta_2}{\sqrt{1-\cos^2\theta_2\cos^2\theta_1}} = 0 \Rightarrow \begin{cases} \cos\theta_2 = 0 \Rightarrow \theta_2 = \pm 90^\circ \\ \sin\theta_1 = 0 \Rightarrow \theta_1 = 0 \end{cases} \tag{38}$$

IV. SIMULATION

In this section, we use real AZ/EL values of a LEO satellite (Fig.3) input to inverse kinematic equation.

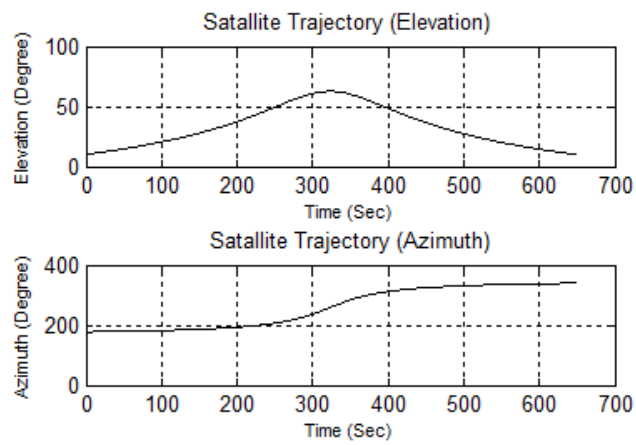

Fig. 3. Satellite path based on real AZ/EL angles.

Then joint angles θ_2 , θ_1 were obtained based on inverse kinematic equation as:

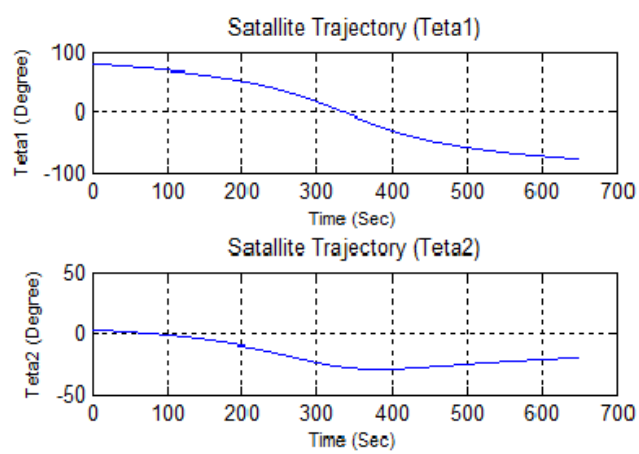

Fig. 4. Joint angles (θ_2 , θ_1)based on real AZ/EL angles of satellite.

As we can see, obtained joint angles θ_2 , θ_1 are in the legal range of X/Y pedestal and do not cover the singular points.

One of the challenging work for further research is to design controller for X/Y servomotors that forward/inverse kinematic

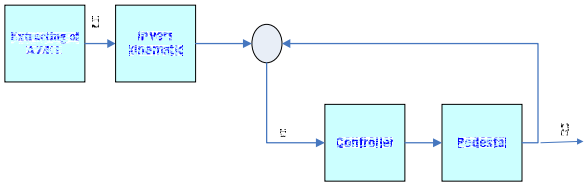

Fig. 5. Block diagram of closed-loop control.

blocks are one of the most important parts. Fig.5 shows our proposed close loop control block diagram.

V. Conclusion

In this paper, we introduce X/Y pedestal with extracting forward/inverse kinematic and dynamic equations also we show the accuracy of inverse kinematic equations. Then we study velocity analysis of each link with extracting singular points, so researchers can continue this case study with introducing a good controller.

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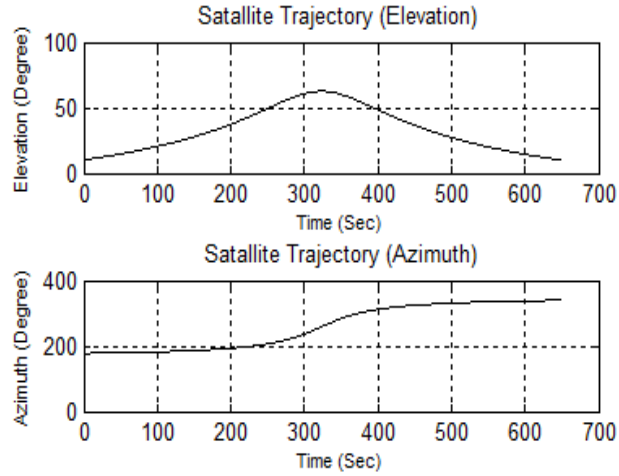
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